EOG Resources - Midland

Loving County, TX (NAD27 TC)
(Scrapfile North)SF-015342_J.B.Walling Svy-A-1392
Scrapfile North #1H

OWB

Design: AWB

Standard Survey Report

11 February, 2019

Company:	EOG Resources - Midland	**Local Co-ordinate Reference:**	Well Scrapfile North #1H
Project:	Loving County, TX (NAD27 TC)	**TVD Reference:**	KB @ 3282.0usft (H&P 646)
Site:	(Scrapfile North)SF-015342_J.B.Walling Svy-A-1392	**MD Reference:**	KB @ 3282.0usft (H&P 646)
Well:	Scrapfile North #1H	**North Reference:**	Grid
Wellbore:	OWB	**Survey Calculation Method:**	Minimum Curvature
Design:	AWB	**Database:**	EDM 10_13

Project Loving County, TX (NAD27 TC)

Map System:	US State Plane 1927 (Exact solution)	**System Datum:**	Mean Sea Level
Geo Datum:	NAD 1927 (NADCON CONUS)		
Map Zone:	Texas Central 4203		

Site (Scrapfile North)SF-015342_J.B.Walling Svy-A-1392

Site Position:		**Northing:**	830,558.00 usft	**Latitude:**	31° 54' 43.149 N
From:	Map	**Easting:**	1,019,428.00 usft	**Longitude:**	103° 29' 37.037 W
Position Uncertainty:	0.0 usft	**Slot Radius:**	13-3/16 "	**Grid Convergence:**	-1.63 °

Well Scrapfile North #1H

Well Position	**+N/-S**	0.0 usft	**Northing:**	830,558.00 usft	**Latitude:**	31° 54' 43.149 N
	+E/-W	0.0 usft	**Easting:**	1,019,428.00 usft	**Longitude:**	103° 29' 37.037 W
Position Uncertainty		0.0 usft	**Wellhead Elevation:**	usft	**Ground Level:**	3,257.0 usft

Wellbore OWB

Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	IGRF2015	01/09/19	6.73	59.77	47,601.59727828

Design AWB

Audit Notes:

Version:	1.0	**Phase:**	ACTUAL	**Tie On Depth:**	0.0

Vertical Section:	Depth From (TVD) (usft)	+N/-S (usft)	+E/-W (usft)	Direction (°)
	0.0	0.0	0.0	1.45

Survey Program **Date** 02/11/19

From (usft)	To (usft)	Survey (Wellbore)	Tool Name	Description
153.0	22,190.0	Intrepid MWD (OWB)	MWD	MWD - Standard

Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
0.0	0.00	0.00	0.0	0.0	0.0	0.0	0.00	0.00	0.00
153.0	0.20	257.80	153.0	-0.1	-0.3	-0.1	0.13	0.13	0.00
248.0	0.40	261.70	248.0	-0.1	-0.8	-0.2	0.21	0.21	4.11
343.0	0.10	221.90	343.0	-0.2	-1.1	-0.3	0.35	-0.32	-41.89
438.0	0.10	284.20	438.0	-0.3	-1.3	-0.3	0.11	0.00	65.58
597.0	0.10	273.70	597.0	-0.2	-1.5	-0.3	0.01	0.00	-6.60
787.0	0.10	355.30	787.0	-0.1	-1.7	-0.1	0.07	0.00	42.95
940.0	0.10	155.30	940.0	-0.1	-1.7	-0.1	0.13	0.00	104.58
1,063.0	0.30	269.00	1,063.0	-0.2	-2.0	-0.2	0.29	0.16	92.44
1,253.0	0.60	228.20	1,253.0	-0.8	-3.2	-0.9	0.22	0.16	-21.47

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Wellbore:	OWB	**Survey Calculation Method:**	Minimum Curvature
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Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
1,443.0	0.50	47.50	1,443.0	-0.9	-3.3	-1.0	0.58	-0.05	94.37
1,633.0	0.60	60.60	1,633.0	0.1	-1.8	0.1	0.08	0.05	6.89
1,824.0	0.40	60.40	1,824.0	0.9	-0.4	0.9	0.10	-0.10	-0.10
2,014.0	0.40	95.40	2,014.0	1.2	0.8	1.2	0.13	0.00	18.42
2,204.0	0.50	53.30	2,204.0	1.6	2.2	1.7	0.18	0.05	-22.16
2,394.0	1.10	46.30	2,393.9	3.4	4.1	3.5	0.32	0.32	-3.68
2,584.0	1.80	39.50	2,583.9	6.9	7.4	7.1	0.38	0.37	-3.58
2,774.0	2.70	21.50	2,773.7	13.4	10.9	13.7	0.60	0.47	-9.47
2,964.0	2.50	11.50	2,963.5	21.6	13.4	22.0	0.26	-0.11	-5.26
3,154.0	2.70	22.70	3,153.3	29.8	15.9	30.2	0.29	0.11	5.89
3,344.0	3.10	23.60	3,343.1	38.7	19.7	39.2	0.21	0.21	0.47
3,534.0	2.70	24.80	3,532.9	47.4	23.6	48.0	0.21	-0.21	0.63
3,724.0	2.20	34.50	3,722.7	54.5	27.6	55.2	0.34	-0.26	5.11
3,915.0	1.70	48.60	3,913.6	59.4	31.8	60.2	0.36	-0.26	7.38
4,104.0	2.60	37.50	4,102.4	64.7	36.5	65.6	0.52	0.48	-5.87
4,294.0	1.70	34.50	4,292.3	70.4	40.7	71.4	0.48	-0.47	-1.58
4,484.0	1.30	37.40	4,482.2	74.4	43.6	75.5	0.21	-0.21	1.53
4,674.0	1.30	73.40	4,672.2	76.8	47.0	77.9	0.42	0.00	18.95
4,864.0	1.70	102.20	4,862.1	76.8	51.8	78.1	0.44	0.21	15.16
5,054.0	3.10	117.00	5,052.0	73.8	59.2	75.3	0.80	0.74	7.79
5,237.0	2.60	118.20	5,234.7	69.6	67.2	71.3	0.28	-0.27	0.66
5,427.0	0.40	195.50	5,424.7	67.0	70.8	68.7	1.34	-1.16	40.68
5,522.0	0.40	186.30	5,519.7	66.3	70.7	68.1	0.07	0.00	-9.68
5,712.0	1.20	279.80	5,709.6	66.0	68.7	67.7	0.68	0.42	49.21
5,902.0	2.00	287.70	5,899.6	67.3	63.6	68.9	0.44	0.42	4.16
6,092.0	2.30	287.10	6,089.4	69.5	56.8	70.9	0.16	0.16	-0.32
6,282.0	2.20	279.80	6,279.3	71.2	49.5	72.4	0.16	-0.05	-3.84
6,471.0	1.80	289.80	6,468.2	72.8	43.2	73.9	0.28	-0.21	5.29
6,661.0	1.80	295.50	6,658.1	75.1	37.7	76.1	0.09	0.00	3.00
6,851.0	1.80	295.40	6,848.0	77.7	32.3	78.5	0.00	0.00	-0.05
7,042.0	0.40	357.50	7,039.0	79.6	29.5	80.4	0.86	-0.73	32.51
7,231.0	0.70	78.20	7,227.9	80.5	30.6	81.3	0.40	0.16	42.70
7,421.0	0.60	128.10	7,417.9	80.2	32.5	81.0	0.29	-0.05	26.26
7,611.0	0.90	215.70	7,607.9	78.3	32.5	79.1	0.56	0.16	46.11
7,801.0	1.50	221.50	7,797.9	75.3	29.9	76.0	0.32	0.32	3.05
7,992.0	0.10	249.70	7,988.9	73.3	28.1	74.0	0.74	-0.73	14.76
8,181.0	0.70	76.90	8,177.9	73.5	29.1	74.3	0.42	0.32	-91.43
8,372.0	0.60	114.30	8,368.8	73.4	31.1	74.2	0.22	-0.05	19.58
8,562.0	0.50	142.30	8,558.8	72.3	32.6	73.1	0.15	-0.05	14.74
8,752.0	0.60	192.50	8,748.8	70.7	32.8	71.5	0.25	0.05	26.42
8,942.0	0.80	196.80	8,938.8	68.5	32.3	69.3	0.11	0.11	2.26
9,132.0	0.90	212.70	9,128.8	65.9	31.1	66.7	0.13	0.05	8.37

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Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
9,323.0	1.00	225.60	9,319.8	63.5	29.1	64.2	0.12	0.05	6.75
9,418.0	1.00	205.70	9,414.8	62.2	28.1	62.9	0.36	0.00	-20.95
9,608.0	0.80	218.80	9,604.7	59.7	26.6	60.3	0.15	-0.11	6.89
9,798.0	1.10	181.90	9,794.7	56.8	25.7	57.4	0.35	0.16	-19.42
9,989.0	0.20	210.80	9,985.7	54.7	25.4	55.3	0.49	-0.47	15.13
10,084.0	0.50	233.20	10,080.7	54.3	25.0	54.9	0.34	0.32	23.58
10,179.0	0.90	202.10	10,175.7	53.3	24.4	53.9	0.57	0.42	-32.74
10,369.0	0.70	258.10	10,365.7	51.7	22.7	52.3	0.41	-0.11	29.47
10,463.0	1.00	261.70	10,459.7	51.5	21.3	52.0	0.32	0.32	3.83
10,653.0	1.70	237.40	10,649.6	49.7	17.3	50.2	0.47	0.37	-12.79
10,844.0	1.40	199.60	10,840.5	46.0	14.2	46.3	0.55	-0.16	-19.79
10,939.0	0.50	186.80	10,935.5	44.5	13.7	44.8	0.97	-0.95	-13.47
11,129.0	0.50	298.80	11,125.5	44.1	12.9	44.4	0.44	0.00	58.95
11,224.0	0.70	54.30	11,220.5	44.6	13.0	44.9	1.07	0.21	121.58
11,414.0	1.00	77.30	11,410.5	45.7	15.6	46.0	0.24	0.16	12.11
11,509.0	0.50	72.50	11,505.5	46.0	16.8	46.4	0.53	-0.53	-5.05
11,641.0	1.00	147.20	11,637.5	45.2	17.9	45.6	0.75	0.38	56.59
11,736.0	7.60	14.90	11,732.2	50.6	20.0	51.0	8.74	6.95	-139.26
11,783.0	13.80	8.10	11,778.4	59.1	21.6	59.6	13.44	13.19	-14.47
11,831.0	15.00	12.70	11,824.9	70.8	23.8	71.4	3.45	2.50	9.58
11,878.0	16.00	15.30	11,870.2	83.0	26.8	83.7	2.59	2.13	5.53
11,878.7	16.08	15.24	11,870.8	83.2	26.9	83.9	11.22	10.99	-8.21
FTP @ 11878.7' MD, 100' FSL, 799' FEL									
11,926.0	21.30	12.30	11,915.6	97.9	30.4	98.7	11.22	11.04	-6.22
11,973.0	29.70	6.40	11,958.0	117.9	33.5	118.7	18.65	17.87	-12.55
12,021.0	34.20	5.50	11,998.7	143.1	36.2	144.0	9.43	9.38	-1.88
12,068.0	39.70	359.00	12,036.3	171.3	37.2	172.2	14.34	11.70	-13.83
12,116.0	47.30	358.80	12,071.1	204.3	36.5	205.2	15.84	15.83	-0.42
12,211.0	60.50	1.50	12,126.9	280.9	36.9	281.8	14.08	13.89	2.84
12,258.0	70.80	1.90	12,146.3	323.7	38.2	324.5	21.93	21.91	0.85
12,306.0	75.40	1.20	12,160.2	369.6	39.4	370.4	9.68	9.58	-1.46
12,353.0	75.20	1.40	12,172.2	415.0	40.4	415.9	0.59	-0.43	0.43
12,401.0	75.00	1.50	12,184.5	461.4	41.6	462.3	0.46	-0.42	0.21
12,448.0	78.60	0.20	12,195.2	507.1	42.3	508.0	8.12	7.66	-2.77
12,496.0	82.80	357.50	12,203.0	554.5	41.3	555.3	10.36	8.75	-5.63
12,543.0	85.10	357.50	12,208.0	601.2	39.3	602.0	4.89	4.89	0.00
12,592.0	85.80	357.00	12,211.8	649.9	36.9	650.7	1.75	1.43	-1.02
12,687.0	86.80	359.30	12,218.0	744.7	33.9	745.3	2.64	1.05	2.42
12,782.0	90.10	2.10	12,220.5	839.6	35.0	840.2	4.55	3.47	2.95
12,877.0	87.80	2.30	12,222.3	934.5	38.7	935.2	2.43	-2.42	0.21
12,972.0	89.80	3.10	12,224.3	1,029.4	43.2	1,030.2	2.27	2.11	0.84
13,067.0	87.50	4.00	12,226.5	1,124.2	49.0	1,125.1	2.60	-2.42	0.95
13,162.0	88.20	3.70	12,230.1	1,218.9	55.4	1,219.9	0.80	0.74	-0.32

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13,257.0	89.00	0.90	12,232.4	1,313.8	59.2	1,314.9	3.06	0.84	-2.95
13,352.0	88.50	359.20	12,234.5	1,408.8	59.3	1,409.8	1.86	-0.53	-1.79
13,447.0	90.20	0.10	12,235.6	1,503.8	58.7	1,504.8	2.02	1.79	0.95
13,542.0	90.90	359.50	12,234.6	1,598.7	58.4	1,599.7	0.97	0.74	-0.63
13,637.0	90.60	358.60	12,233.4	1,693.7	56.8	1,694.6	1.00	-0.32	-0.95
13,732.0	87.70	0.10	12,234.8	1,788.7	55.7	1,789.5	3.44	-3.05	1.58
13,827.0	89.10	0.00	12,237.5	1,883.7	55.8	1,884.5	1.48	1.47	-0.11
13,922.0	90.70	0.40	12,237.6	1,978.7	56.2	1,979.4	1.74	1.68	0.42
14,007.0	87.80	1.20	12,238.7	2,063.6	57.3	2,064.4	3.54	-3.41	0.94
14,102.0	89.10	0.40	12,241.3	2,158.6	58.7	2,159.4	1.61	1.37	-0.84
14,197.0	90.30	0.80	12,241.8	2,253.6	59.7	2,254.4	1.33	1.26	0.42
14,292.0	85.90	1.60	12,245.0	2,348.5	61.6	2,349.3	4.71	-4.63	0.84
14,387.0	86.20	1.20	12,251.5	2,443.2	64.0	2,444.1	0.53	0.32	-0.42
14,482.0	88.70	1.70	12,255.7	2,538.1	66.4	2,539.0	2.68	2.63	0.53
14,577.0	88.50	359.90	12,258.0	2,633.0	67.7	2,633.9	1.91	-0.21	-1.89
14,673.0	89.70	358.20	12,259.6	2,729.0	66.1	2,729.8	2.17	1.25	-1.77
14,768.0	89.10	356.70	12,260.5	2,823.9	61.9	2,824.6	1.70	-0.63	-1.58
14,863.0	90.50	358.70	12,260.9	2,918.8	58.1	2,919.4	2.57	1.47	2.11
14,958.0	88.60	2.50	12,261.6	3,013.8	59.1	3,014.3	4.47	-2.00	4.00
15,053.0	88.10	1.40	12,264.4	3,108.7	62.3	3,109.3	1.27	-0.53	-1.16
15,148.0	91.30	2.30	12,264.9	3,203.6	65.4	3,204.3	3.50	3.37	0.95
15,243.0	93.30	4.80	12,261.0	3,298.4	71.2	3,299.1	3.37	2.11	2.63
15,338.0	92.90	4.70	12,255.9	3,392.9	79.1	3,393.8	0.43	-0.42	-0.11
15,433.0	88.70	1.50	12,254.6	3,487.7	84.2	3,488.7	5.56	-4.42	-3.37
15,528.0	89.40	0.20	12,256.2	3,582.7	85.6	3,583.7	1.55	0.74	-1.37
15,623.0	88.80	0.00	12,257.6	3,677.7	85.8	3,678.7	0.67	-0.63	-0.21
15,718.0	87.50	0.30	12,260.7	3,772.6	86.0	3,773.6	1.40	-1.37	0.32
15,813.0	87.20	359.60	12,265.1	3,867.5	86.0	3,868.5	0.80	-0.32	-0.74
15,908.0	87.30	359.50	12,269.7	3,962.4	85.2	3,963.3	0.15	0.11	-0.11
16,003.0	91.40	359.10	12,270.7	4,057.4	84.1	4,058.2	4.34	4.32	-0.42
16,098.0	91.30	0.40	12,268.5	4,152.4	83.6	4,153.1	1.37	-0.11	1.37
16,193.0	91.30	2.80	12,266.3	4,247.3	86.3	4,248.1	2.53	0.00	2.53
16,288.0	89.60	4.90	12,265.6	4,342.1	92.7	4,343.0	2.84	-1.79	2.21
16,383.0	87.10	4.80	12,268.3	4,436.7	100.7	4,437.8	2.63	-2.63	-0.11
16,478.0	88.00	6.10	12,272.4	4,531.2	109.7	4,532.5	1.66	0.95	1.37
16,573.0	91.40	3.90	12,272.9	4,625.8	118.0	4,627.3	4.26	3.58	-2.32
16,668.0	89.70	3.30	12,272.0	4,720.6	124.0	4,722.2	1.90	-1.79	-0.63
16,763.0	87.80	4.50	12,274.1	4,815.3	130.4	4,817.1	2.37	-2.00	1.26
16,858.0	86.90	2.80	12,278.5	4,910.0	136.5	4,911.9	2.02	-0.95	-1.79
16,953.0	88.00	2.20	12,282.7	5,004.8	140.6	5,006.8	1.32	1.16	-0.63
17,048.0	91.60	1.50	12,283.0	5,099.8	143.7	5,101.8	3.86	3.79	-0.74
17,143.0	90.10	0.30	12,281.6	5,194.7	145.1	5,196.8	2.02	-1.58	-1.26

Company:	EOG Resources - Midland	**Local Co-ordinate Reference:**	Well Scrapfile North #1H
Project:	Loving County, TX (NAD27 TC)	**TVD Reference:**	KB @ 3282.0usft (H&P 646)
Site:	(Scrapfile North)SF-015342_J.B.Walling Svy-A-1392	**MD Reference:**	KB @ 3282.0usft (H&P 646)
Well:	Scrapfile North #1H	**North Reference:**	Grid
Wellbore:	OWB	**Survey Calculation Method:**	Minimum Curvature
Design:	AWB	**Database:**	EDM 10_13

Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
17,238.0	88.40	3.50	12,282.8	5,289.7	148.3	5,291.7	3.81	-1.79	3.37
17,333.0	87.20	1.50	12,286.5	5,384.5	152.4	5,386.6	2.45	-1.26	-2.11
17,428.0	84.10	1.00	12,293.7	5,479.2	154.5	5,481.4	3.31	-3.26	-0.53
17,523.0	85.50	0.90	12,302.3	5,573.8	156.1	5,576.0	1.48	1.47	-0.11
17,618.0	87.60	357.90	12,308.0	5,668.6	155.1	5,670.7	3.85	2.21	-3.16
17,713.0	89.20	359.40	12,310.7	5,763.5	152.8	5,765.5	2.31	1.68	1.58
17,808.0	89.50	358.10	12,311.8	5,858.5	150.8	5,860.4	1.40	0.32	-1.37
17,903.0	89.50	359.10	12,312.6	5,953.5	148.4	5,955.3	1.05	0.00	1.05
17,999.0	88.80	357.60	12,314.0	6,049.4	145.7	6,051.2	1.72	-0.73	-1.56
18,093.0	90.00	358.80	12,315.0	6,143.4	142.7	6,145.0	1.81	1.28	1.28
18,188.0	89.90	358.20	12,315.1	6,238.3	140.2	6,239.9	0.64	-0.11	-0.63
18,283.0	89.30	0.40	12,315.7	6,333.3	139.1	6,334.8	2.40	-0.63	2.32
18,378.0	87.90	2.60	12,318.1	6,428.2	141.6	6,429.8	2.74	-1.47	2.32
18,473.0	88.20	5.20	12,321.3	6,523.0	148.0	6,524.6	2.75	0.32	2.74
18,569.0	89.10	5.40	12,323.6	6,618.5	156.9	6,620.4	0.96	0.94	0.21
18,664.0	86.80	1.80	12,327.0	6,713.2	162.9	6,715.2	4.49	-2.42	-3.79
18,759.0	88.40	0.80	12,330.9	6,808.1	165.0	6,810.1	1.99	1.68	-1.05
18,854.0	92.10	0.40	12,330.5	6,903.1	166.0	6,905.1	3.92	3.89	-0.42
18,949.0	90.50	3.10	12,328.4	6,998.0	168.9	7,000.1	3.30	-1.68	2.84
19,044.0	87.60	5.90	12,329.9	7,092.7	176.4	7,094.9	4.24	-3.05	2.95
19,138.0	87.60	4.90	12,333.9	7,186.2	185.2	7,188.6	1.06	0.00	-1.06
19,233.0	86.10	3.30	12,339.1	7,280.8	192.0	7,283.3	2.31	-1.58	-1.68
19,328.0	87.20	3.30	12,344.6	7,375.5	197.4	7,378.1	1.16	1.16	0.00
19,423.0	88.10	4.70	12,348.5	7,470.2	204.1	7,472.9	1.75	0.95	1.47
19,517.0	88.50	2.60	12,351.3	7,563.9	210.0	7,566.8	2.27	0.43	-2.23
19,612.0	88.60	4.90	12,353.7	7,658.7	216.2	7,661.7	2.42	0.11	2.42
19,707.0	88.40	4.90	12,356.2	7,753.3	224.4	7,756.5	0.21	-0.21	0.00
19,802.0	86.90	4.70	12,360.1	7,847.9	232.3	7,851.3	1.59	-1.58	-0.21
19,896.0	86.50	0.20	12,365.5	7,941.6	236.3	7,945.1	4.80	-0.43	-4.79
19,991.0	90.40	359.90	12,368.1	8,036.6	236.4	8,040.0	4.12	4.11	-0.32
20,086.0	89.00	358.40	12,368.6	8,131.6	235.0	8,134.9	2.16	-1.47	-1.58
20,182.0	89.90	358.50	12,369.5	8,227.5	232.4	8,230.8	0.94	0.94	0.10
20,276.0	91.30	358.40	12,368.5	8,321.5	229.8	8,324.6	1.49	1.49	-0.11
20,371.0	89.10	358.50	12,368.2	8,416.4	227.3	8,419.5	2.32	-2.32	0.11
20,466.0	89.30	0.10	12,369.5	8,511.4	226.1	8,514.4	1.70	0.21	1.68
20,561.0	88.40	0.10	12,371.4	8,606.4	226.3	8,609.4	0.95	-0.95	0.00
20,656.0	89.40	4.70	12,373.3	8,701.3	230.3	8,704.3	4.95	1.05	4.84
20,751.0	87.60	3.50	12,375.7	8,796.0	237.1	8,799.2	2.28	-1.89	-1.26
20,846.0	87.20	358.90	12,380.1	8,890.9	239.0	8,894.1	4.86	-0.42	-4.84
20,942.0	87.80	359.10	12,384.2	8,986.7	237.4	8,989.9	0.66	0.63	0.21
21,037.0	88.00	0.20	12,387.7	9,081.7	236.8	9,084.8	1.18	0.21	1.16
21,132.0	88.60	2.40	12,390.5	9,176.6	238.9	9,179.7	2.40	0.63	2.32
21,227.0	88.70	359.40	12,392.8	9,271.6	240.4	9,274.7	3.16	0.11	-3.16

Company:	EOG Resources - Midland	**Local Co-ordinate Reference:**	Well Scrapfile North #1H
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Site:	(Scrapfile North)SF-015342_J.B.Walling Svy-A-1392	**MD Reference:**	KB @ 3282.0usft (H&P 646)
Well:	Scrapfile North #1H	**North Reference:**	Grid
Wellbore:	OWB	**Survey Calculation Method:**	Minimum Curvature
Design:	AWB	**Database:**	EDM 10_13

Survey

Measured Depth (usft)	Inclination (°)	Azimuth (°)	Vertical Depth (usft)	+N/-S (usft)	+E/-W (usft)	Vertical Section (usft)	Dogleg Rate (°/100usft)	Build Rate (°/100usft)	Turn Rate (°/100usft)
21,322.0	89.70	1.10	12,394.1	9,366.5	240.8	9,369.6	2.08	1.05	1.79
21,417.0	89.50	0.70	12,394.8	9,461.5	242.3	9,464.6	0.47	-0.21	-0.42
21,512.0	89.40	1.60	12,395.7	9,556.5	244.2	9,559.6	0.95	-0.11	0.95
21,607.0	89.90	1.80	12,396.3	9,651.5	247.1	9,654.6	0.57	0.53	0.21
21,702.0	90.50	359.80	12,395.9	9,746.4	248.4	9,749.6	2.20	0.63	-2.11
21,796.0	90.30	0.60	12,395.3	9,840.4	248.7	9,843.6	0.88	-0.21	0.85
21,892.0	90.20	359.60	12,394.9	9,936.4	248.9	9,939.6	1.05	-0.10	-1.04
21,986.0	86.50	359.10	12,397.6	10,030.4	247.8	10,033.4	3.97	-3.94	-0.53
22,081.0	86.60	359.60	12,403.3	10,125.2	246.7	10,128.2	0.54	0.11	0.53
22,126.0	86.60	359.60	12,405.9	10,170.1	246.4	10,173.1	0.00	0.00	0.00
LAST SVY									
22,190.0	86.60	359.60	12,409.7	10,234.0	246.0	10,236.9	0.00	0.00	0.00
PTD @ 112' FNL, 857' FEL									

Design Targets

Target Name - hit/miss target - Shape	Dip Angle (°)	Dip Dir. (°)	TVD (usft)	+N/-S (usft)	+E/-W (usft)	Northing (usft)	Easting (usft)	Latitude	Longitude
POP (Scrapfile North	0.00	360.00	11,732.6	58.0	19.0	830,616.00	1,019,447.00	31° 54' 43.728 N	103° 29' 36.836 W
- actual wellpath misses target center by 7.4usft at 11737.7usft MD (11733.9 TVD, 50.8 N, 20.1 E)									
- Rectangle (sides W100.0 H50.0 D0.0)									
FTP (Scrapfile North #	0.00	360.00	12,210.0	83.0	21.0	830,641.00	1,019,449.00	31° 54' 43.976 N	103° 29' 36.821 W
- actual wellpath misses target center by 185.1usft at 12116.0usft MD (12071.1 TVD, 204.3 N, 36.5 E)									
- Point									
TGT 1 (Scrapfile Nortl	0.00	0.00	12,253.0	2,498.9	76.7	833,056.90	1,019,504.75	31° 55' 7.889 N	103° 29' 36.970 W
- actual wellpath misses target center by 11.6usft at 14443.0usft MD (12254.5 TVD, 2499.1 N, 65.3 E)									
- Point									
TGT 2 (Scrapfile Nortl	0.00	360.00	12,300.0	4,798.2	131.1	835,356.20	1,019,559.14	31° 55' 30.649 N	103° 29' 37.097 W
- actual wellpath misses target center by 26.6usft at 16746.5usft MD (12273.5 TVD, 4798.9 N, 129.1 E)									
- Point									
TGT 3 (Scrapfile Nortl	0.00	0.00	12,342.0	7,397.5	192.6	837,955.50	1,019,620.64	31° 55' 56.378 N	103° 29' 37.241 W
- actual wellpath misses target center by 7.1usft at 19349.5usft MD (12345.7 TVD, 7396.9 N, 198.7 E)									
- Point									
PBHL (Scrapfile North	0.00	181.36	12,386.0	10,245.0	260.0	840,803.00	1,019,688.00	31° 56' 24.564 N	103° 29' 37.398 W
- actual wellpath misses target center by 29.7usft at 22190.0usft MD (12409.7 TVD, 10234.0 N, 246.0 E)									
- Rectangle (sides W60.0 H10,140.0 D0.0)									

Design Annotations

Measured Depth (usft)	Vertical Depth (usft)	Local Coordinates +N/-S (usft)	Local Coordinates +E/-W (usft)	Comment
11,878.7	11,870.8	83.2	26.9	FTP @ 11878.7' MD, 100' FSL, 799' FEL
22,126.0	12,405.9	10,170.1	246.4	LAST SVY
22,190.0	12,409.7	10,234.0	246.0	PTD @ 112' FNL, 857' FEL

Company:	EOG Resources - Midland	**Local Co-ordinate Reference:**	Well Scrapfile North #1H
Project:	Loving County, TX (NAD27 TC)	**TVD Reference:**	KB @ 3282.0usft (H&P 646)
Site:	(Scrapfile North)SF-015342_J.B.Walling Svy-A-1392	**MD Reference:**	KB @ 3282.0usft (H&P 646)
Well:	Scrapfile North #1H	**North Reference:**	Grid
Wellbore:	OWB	**Survey Calculation Method:**	Minimum Curvature
Design:	AWB	**Database:**	EDM 10_13

Checked By: _____ Approved By: _____ Date: _____